

SINAN, China

WMO: 577310

Lat: 27.9311N

Lon: 108.2450E

Elev: 555

StdP: 94.83

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.6	1.9	-5.7	2.5	4.9	-3.9	2.9	4.1	3.7	8.5	3.3	8.2	1.0	0	0.311

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.8	35.6	24.5	34.5	24.4	33.3	24.2	26.1	32.1	25.6	31.4	25.2	30.6	2.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.7	21.1	28.4	24.3	20.5	27.9	23.9	20.0	27.7	85.1	32.3	83.0	31.5	81.0	30.7	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
4.3	3.5	3.1	DB	0.0	37.5	1.3	1.0	-0.9	38.2	-1.6	38.8	-2.3	39.3	-3.2	40.0	
			WB	-1.3	26.9	1.3	0.7	-2.2	27.3	-3.0	27.7	-3.8	28.1	-4.7	28.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	18.3	6.9	9.1	13.3	18.5	22.3	25.2	28.3	28.1	24.6	19.1	14.2	8.8	(6)
		DBStd	8.00	3.21	3.89	4.05	3.86	2.86	2.72	2.33	2.51	3.34	3.20	3.24	2.64	
		HDD10.0	235	105	58	13	1	0	0	0	0	0	5	53		
		HDD18.3	1276	354	259	163	44	2	0	0	1	29	129	295		
		CDD10.0	3249	9	33	117	257	382	456	566	438	282	131	17		
		CDD18.3	1247	0	1	8	49	126	206	308	303	189	52	0		
		CDH23.3	11411	1	9	56	319	795	1648	3467	3297	1567	236	16	0	
(13)		CDH26.7	4798	0	2	13	94	268	606	1602	1531	624	56	2	0	(13)
(14)	Wind	WSAvg	1.2	1.0	1.1	1.2	1.2	1.2	1.2	1.5	1.4	1.2	1.0	1.0	0.9	(14)
(15)	Precipitation	PrecAvg	1137	27	29	55	109	166	200	179	120	80	99	50	24	(15)
(16)		PrecMax	1417	72	68	133	213	228	347	438	261	153	193	118	71	(16)
(17)		PrecMin	828	6	2	19	29	82	75	25	33	6	45	6	1	(17)
(18)		PrecStd	148	15	17	23	47	40	72	104	62	34	40	28	16	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	18.0	24.5	28.4	31.8	33.6	34.7	36.8	37.3	35.7	31.1	25.3	17.7		(19)
(20)		0.4% MCWB	12.0	15.9	18.9	21.3	22.3	24.6	24.8	24.3	23.7	21.7	18.5	12.4		(20)
(21)		2% DB	14.7	19.3	24.6	29.4	31.7	33.2	35.5	35.8	33.7	28.1	22.6	15.6		(21)
(22)		2% MCWB	10.1	13.5	17.3	20.4	21.5	24.1	24.9	24.4	23.5	20.0	16.6	11.3		(22)
(23)		5% DB	12.6	16.4	21.8	27.1	29.7	31.8	34.4	34.5	32.2	26.0	20.4	13.8		(23)
(24)		5% MCWB	9.0	11.8	15.4	19.6	21.4	24.0	24.8	24.4	23.1	19.5	15.8	10.1		(24)
(25)		10% DB	11.2	14.2	19.3	24.7	27.7	30.3	33.1	33.2	30.4	23.9	18.6	12.5		(25)
(26)		10% MCWB	8.2	10.7	14.4	18.7	21.0	23.8	24.7	24.3	22.6	18.9	15.1	9.4		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	13.3	16.3	19.9	23.0	24.8	26.1	26.9	26.6	25.4	22.9	19.5	14.0		(27)
(28)		0.4% MCDB	17.1	23.7	27.2	29.0	30.2	32.0	32.8	32.9	32.0	28.3	24.6	16.6		(28)
(29)		2% WB	10.7	14.2	17.9	21.7	23.7	25.5	26.2	26.1	24.7	21.6	17.7	12.0		(29)
(30)		2% MCDB	13.3	18.6	23.4	27.0	28.5	30.8	31.9	32.4	30.7	25.7	20.7	14.3		(30)
(31)		5% WB	9.6	12.5	16.4	20.5	22.8	24.9	25.7	25.6	24.1	20.5	16.7	10.9		(31)
(32)		5% MCDB	12.0	15.6	20.6	25.3	27.3	29.5	31.3	31.4	29.7	24.2	19.5	13.3		(32)
(33)		10% WB	8.6	11.2	15.0	19.4	21.9	24.4	25.3	25.1	23.5	19.5	15.5	9.8		(33)
(34)		10% MCDB	11.0	13.9	18.6	23.7	25.9	28.6	30.7	30.4	28.7	22.8	18.1	12.0		(34)
(35)	Mean Daily Temperature Range	MDBR	4.2	5.2	6.5	7.4	7.4	6.7	7.8	8.3	7.5	5.7	5.3	4.4		(35)
(36)		5% DB MCDBR	8.1	10.4	11.9	12.6	12.2	10.0	10.2	10.7	10.7	9.9	9.3	7.7		(36)
(37)		5% DB MCWBR	4.3	5.1	5.1	4.7	3.7	2.8	2.1	2.2	2.5	3.2	4.0	4.0		(37)
(38)		5% WB MCDBR	7.4	9.5	10.3	10.2	8.4	7.6	8.7	9.1	9.0	7.6	7.6	6.5		(38)
(39)		5% WB MCWBR	4.2	5.0	4.9	4.6	3.5	2.6	2.4	2.4	2.5	3.2	3.6	3.7		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.654	0.668	0.730	0.698	0.603	0.579	0.527	0.574	0.602	0.682	0.673	0.665		(40)
(41)		taud	1.520	1.525	1.462	1.568	1.813	1.910	2.069	1.914	1.803	1.571	1.544	1.509		(41)
(42)		Ebn at Noon	571	617	615	659	727	741	779	740	701	598	554	536		(42)
(43)		Edh at Noon	249	266	299	277	217	196	166	193	210	250	240	242		(43)
(44)	All-Sky Solar Radiation	RadAvg	1.43	1.86	2.61	3.21	3.47	3.52	4.71	4.39	3.45	2.31	1.95	1.47		(44)
(45)		RadStd	0.33	0.45	0.33	0.51	0.61	0.57	0.45	0.45	0.45	0.39	0.39	0.33		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.30	N/A	N/A	+0.51	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	+0.60	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page